Portfolio Part 1 – Conception Phase

Project idea and audience

This project explains how AWS can host a simple webpage and how an active-passive design could respond to a regional application failure. It is intended for a learner or reviewer who needs to understand the request path and the role of each cloud component without first deploying a production system. The deliverables are a responsive interactive explainer, a conceptual architecture diagram, and Terraform source.

Proposed architecture and rationale

The actual demonstration website is a static page stored in Amazon S3 and delivered through Amazon CloudFront. CloudFront uses Origin Access Control (OAC) to read a private S3 bucket, so visitors do not need direct bucket access. HTTPS is enforced at the viewer edge. Terraform describes these resources to make the hosting setup reviewable and repeatable.

Separately, the explainer presents a proposed active-passive application design. During normal operation, CloudFront forwards requests to an Application Load Balancer (ALB) in a primary AWS region. The ALB distributes traffic to EC2 instances across two Availability Zones, and an Auto Scaling group adjusts compute capacity to demand. A secondary region holds a maintenance response. CloudFront origin-group failover can select that response for configured primary-origin failures. Route 53 health checks and failover records are a separate proposed DNS layer; they do not share the same trigger or behavior as CloudFront origin failover.

The component choices separate global entry and caching (CloudFront), private object storage (S3), regional traffic distribution (ALB), and elastic compute (EC2 Auto Scaling). A static standby lowers idle compute cost, but it is not a warm replacement application and cannot serve stateful user workloads.

Objectives and success criteria

The design aims to explain the normal and failure request paths, provide a responsive webpage, host the real explainer privately behind HTTPS, and distinguish implemented resources from proposed behavior. A reader should be able to identify the purpose of S3, CloudFront, Route 53, ALB, and EC2, and understand which mechanisms are active in each scenario.

Scope, assumptions, and constraints

The live deployment is limited to the static explainer's private S3 bucket, CloudFront distribution, OAC, and access policy. The browser scenario controls are visual simulations; they do not call AWS or cause failover. The separate multi-region Terraform root is a prototype for CloudFront origin failover. It does not configure Route 53 records, public health checks, a custom domain, or a custom HTTPS certificate. DNS failover would require a controlled domain and hosted zone, reachable health-check targets, compatible HTTPS on both targets, additional permissions, cost review, and runtime testing. The secondary S3 website endpoint is public for its maintenance object and HTTP-only; it must contain no sensitive information.

Availability and low latency are design objectives, not measured results or AWS service guarantees. CloudFront propagation, regional quotas, account permissions, service charges, DNS caching, and the gap between conceptual design and deployed resources are explicit project constraints. The accompanying diagram is conceptual, not a deployed-resource inventory.

Architecture illustration

Conceptual design only. This diagram is not an inventory of deployed AWS resources.

